Structure – Activity Relationship in e-lysine peptides: The Length Effects on Antifungal Activity

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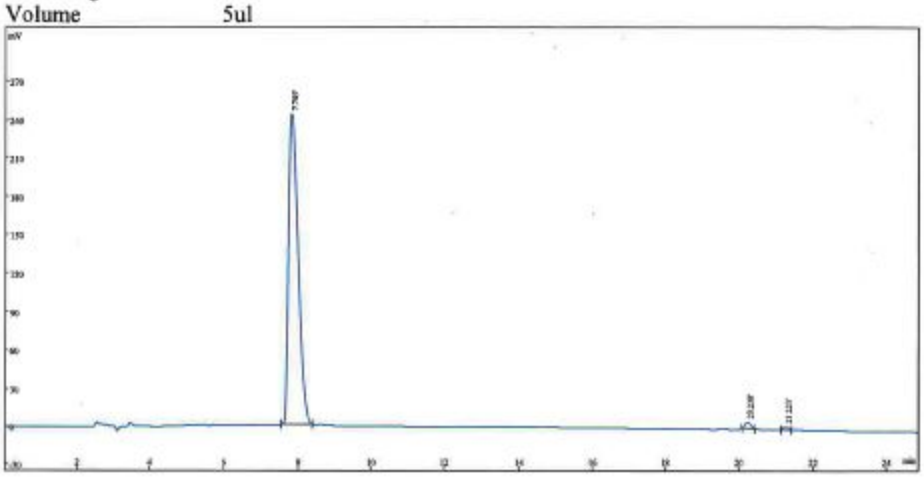
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Fig. S1

Product Name	3269801
Lot No	P3269801-JQ068545
Column	4.6×250mm, YMC-Triart C18
Solvent A	0.1% trifluoroacetic in 100% acetonitrile
Solvent B	0.1% trifluoroacetic in 100% water
Gradient	A B
	0.01min 1% 99%
	25min 26% 74%
	25.1min 100% 0%
	30min STOP
Flow rate	1.0 mL/min
Wavelength	214nm
Volume	5ul



Rank	Time	Conc.	Area	Height
1	7.785	98.0350	3985653	242136
2	20.238	1.4457	58777	5215
3	21.223	0.5193	21112	2161
Total		100	4065542	249512

Fig S1. RP-HPLC of homopeptide containing 12 ε-lysine residues.

Fig. S2

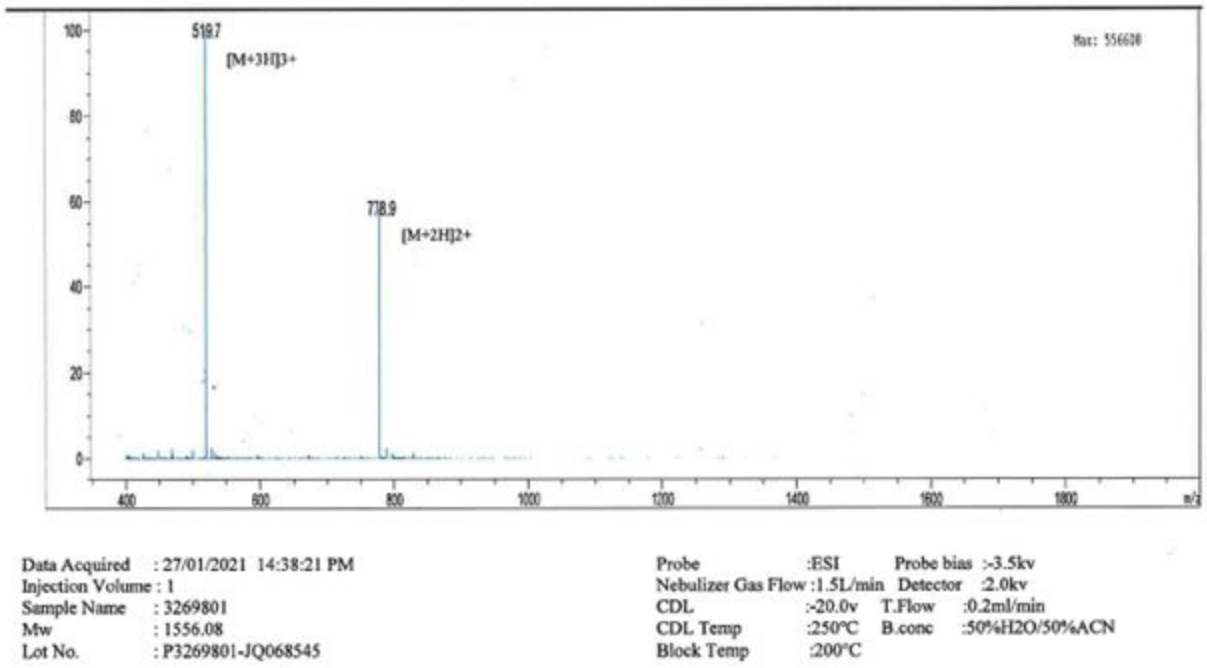
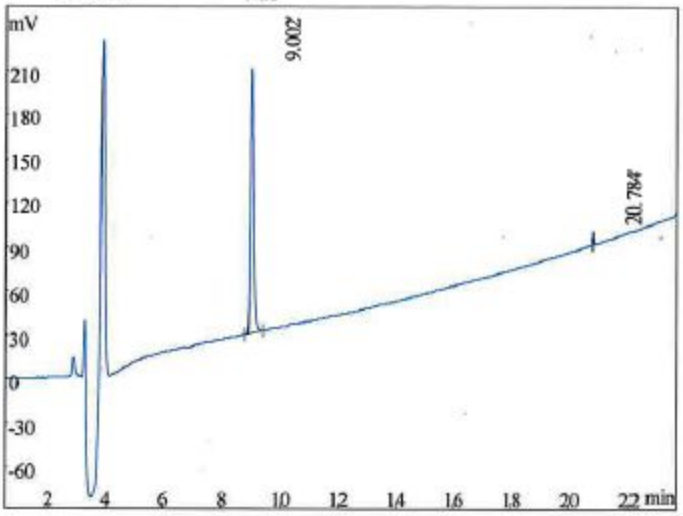


Fig S2. LC-MS of homopeptide containing 12 ε-lysine residues.

Fig. S3

Product Name	3269802
Lot No	P3269802-JQ403999
Column	4.6×250mm, Kromasil 100-5C18
Solvent A	0.1% trifluoroacetic in 100% acetonitrile
Solvent B	0.1% trifluoroacetic in 100% water
Gradient	A B
	0.01min 10% 90%
	25min 67% 33%
	25.1min 100% 0%
	30min STOP
Flow rate	1.0 mL/min
Wavelength	214nm
Volume	5ul



Rank	Time	Conc.	Area	Height
1	9.002	98.34	1174600	179949
2	20.784	1.661	19836	9308
Total		100	1194436	189257

Fig S3. RP-HPLC of homopeptide containing 14 ε-lysine residues.

Fig. S4

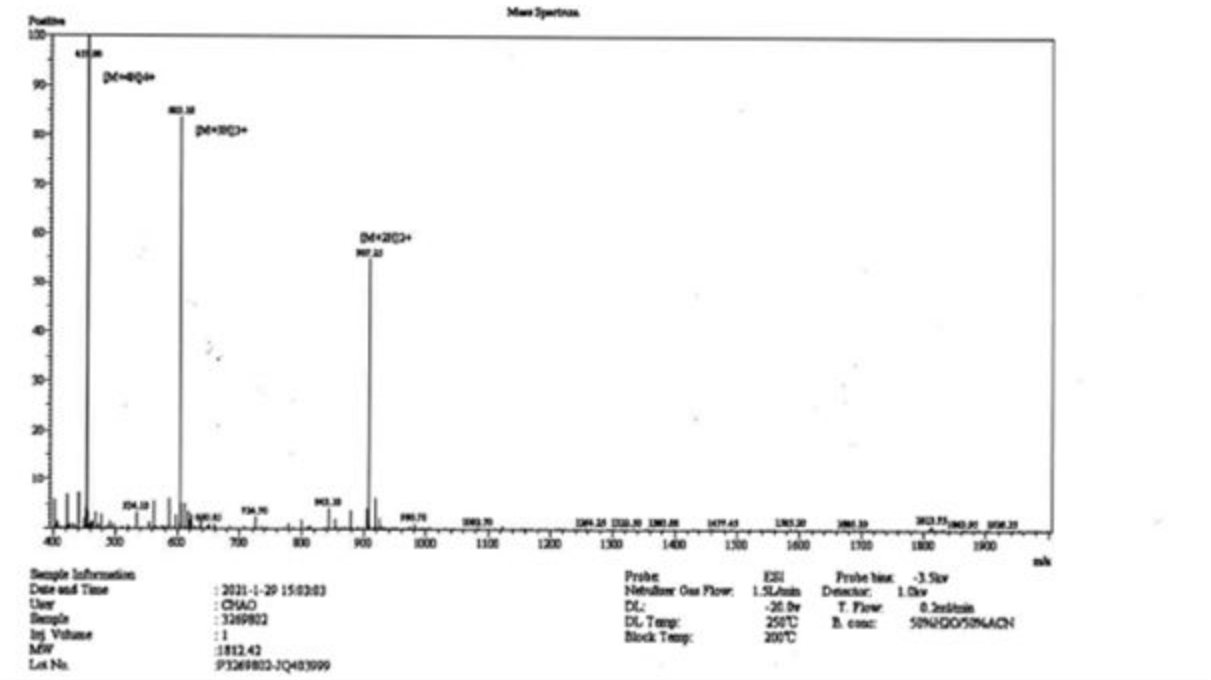


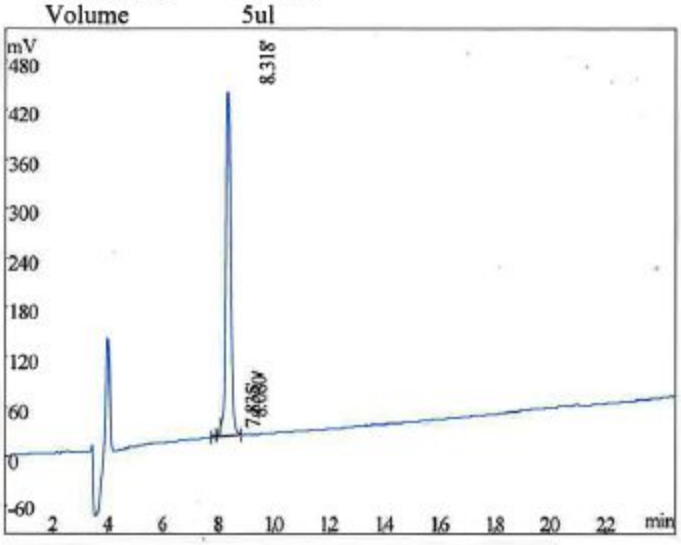
Fig S4. LC-MS of homopeptide containing 14 ε-lysine residues.

Fig. S5

Product Name	3269803
Lot No	P3269803-JQ109574
Column	4.6×250mm, SinoChrom ODS-BP
Solvent A	0.1% trifluoroacetic in 100% acetonitrile
Solvent B	0.1% trifluoroacetic in 100% water

Gradient	A	B
0.01min	2%	98%
25min	59%	41%
25.1min	100%	0%
30min	STOP	

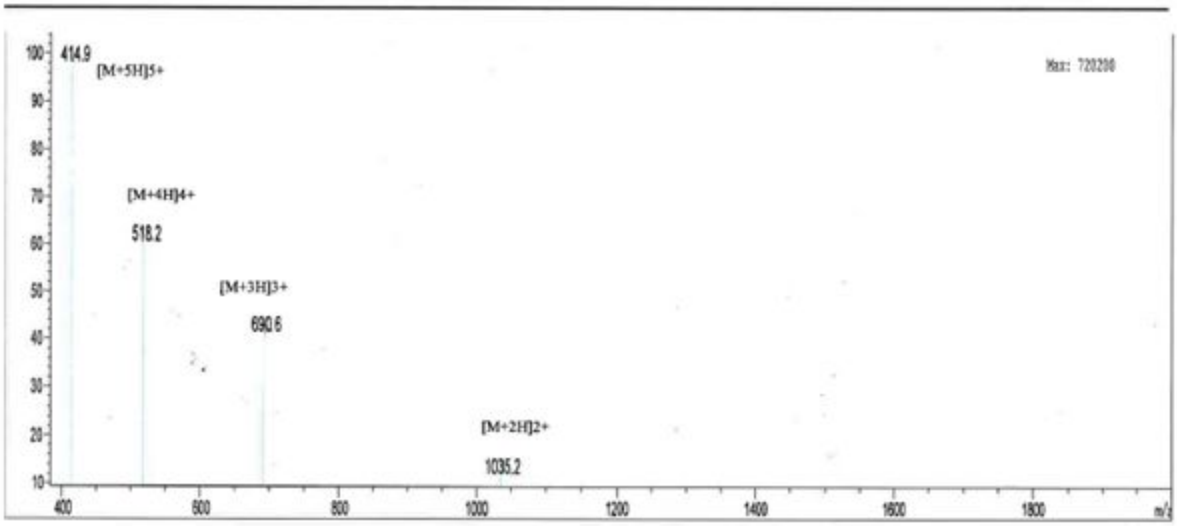
Flow rate	1.0 mL/min
Wavelength	214nm
Volume	5ul



Rank	Time	Conc.	Area	Height
1	7.835	0.5321	23950	2914
2	8.080	1.244	55972	14710
3	8.318	98.22	4421121	416721
Total		100	4501043	434345

Fig S5. RP-HPLC of homopeptide containing 16 ε-lysine residues.

Fig. S6



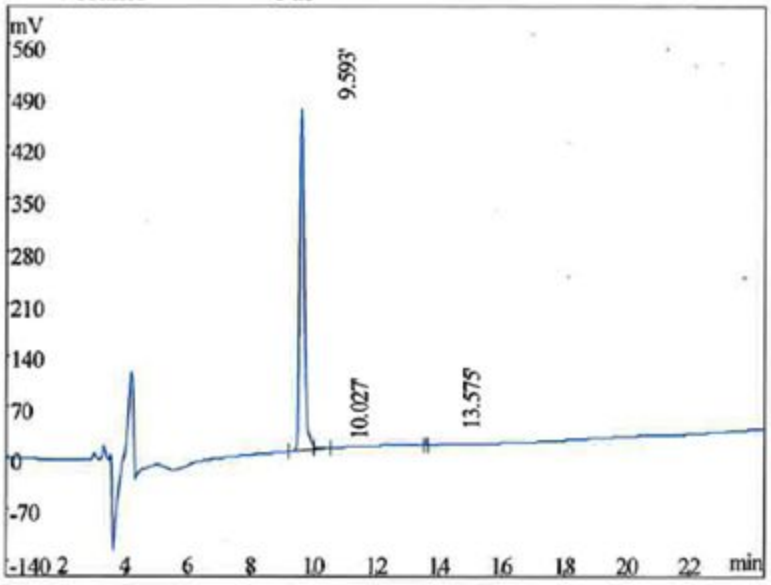
Data Acquired : 02/02/2021 10:45:08 AM
Injection Volume : 1
Sample Name : 3269803
Mw : 2068.77
Lot No. : P3269803-JQ109574

Probe :ESI Probe bias :-3.5kv
Nebulizer Gas Flow :1.5L/min Detector :2.0kv
CDL :-20.0v T.Flow :0.2ml/min
CDL Temp :250°C B.conc :50%H2O/50%ACN
Block Temp :200°C

Fig S6. LC-MS of homopeptide containing 16 ε-lysine residues.

Fig. S7

Product Name	3269804		
Lot No	P3269804-JQ058011		
Column	4.6×250mm, Kromasil 100-5C18		
Solvent A	0.1% trifluoroacetic in 100% acetonitrile		
Solvent B	0.1% trifluoroacetic in 100% water		
Gradient		A	B
	0.01min	10%	90%
	25min	67%	33%
	25.1min	100%	0%
	30min	STOP	
Flow rate	1.0 mL/min		
Wavelength	214nm		
Volume	5ul		



Rank	Time	Conc.	Area	Height
1	9.593	98.6	4190794	463690
2	10.027	0.9169	38970	2751
3	13.575	0.4823	20498	9430
Total		100	4250262	475871

Fig S7. RP-HPLC of homopeptide containing 18 ε-lysine residues.

Fig. S8

Fig S8. LC-MS of homopeptide containing 18 ε-lysine residues.

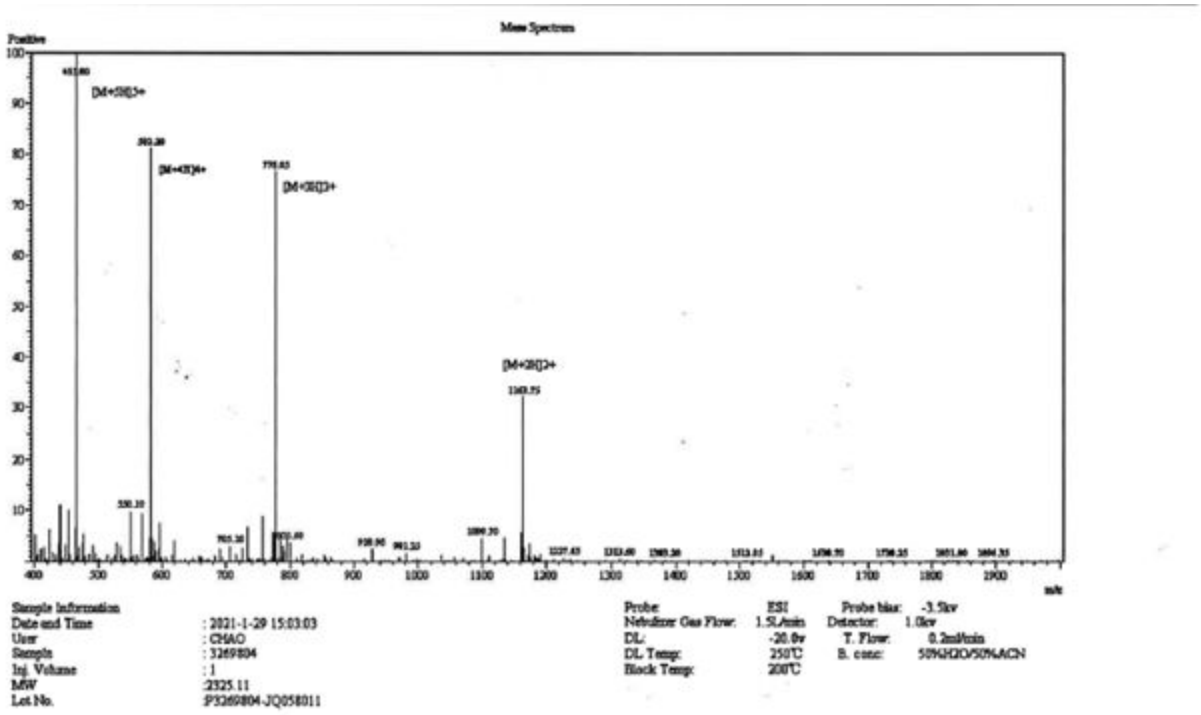


Fig. S9

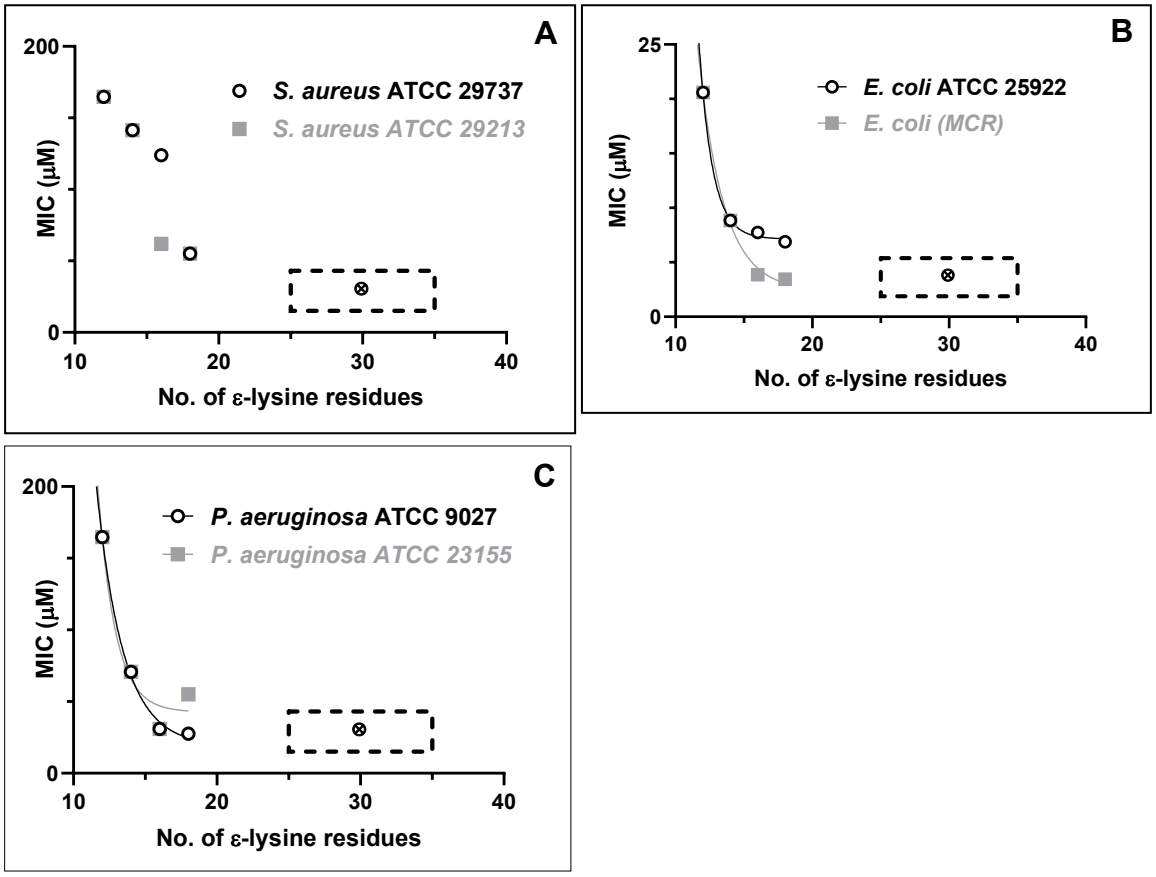


Fig. S9 Changes in the minimum inhibitory concentrations (MIC in μM) of εPL peptides and εPL polymer against **A**) Gram-positive and **B,C**) Gram-negative bacteria.

Fig. S10

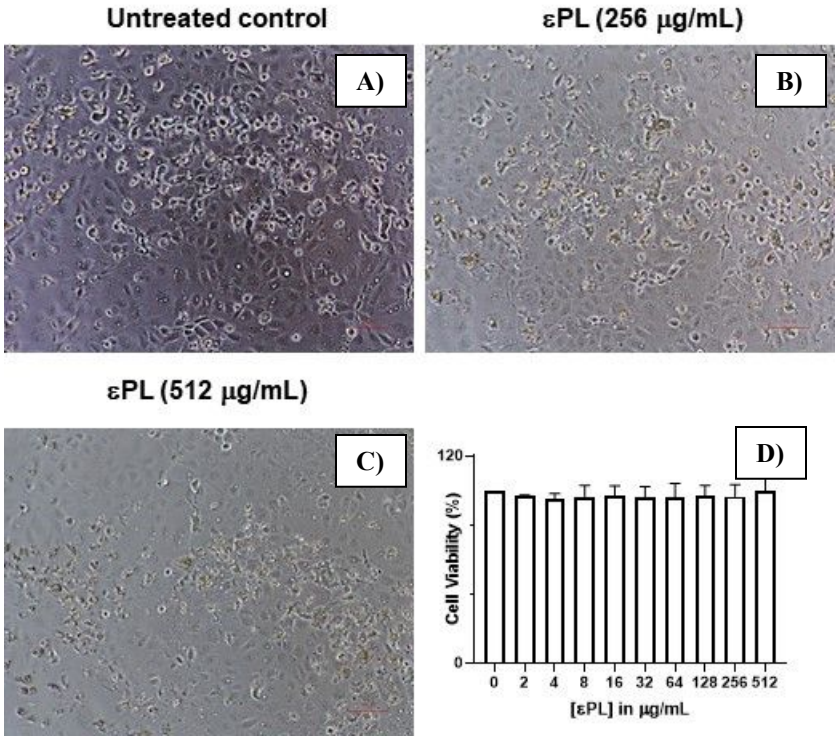


Fig. S10 Optical micrographs showing the morphology of HCET cells treated with various groups. **A**) Untreated control; **B**) εPL (256 μg/mL); **C**) εPL (512 μg/mL) and **d**) Metabolic activity of HCET cells determined by WST-1 assay.

Fig. S11

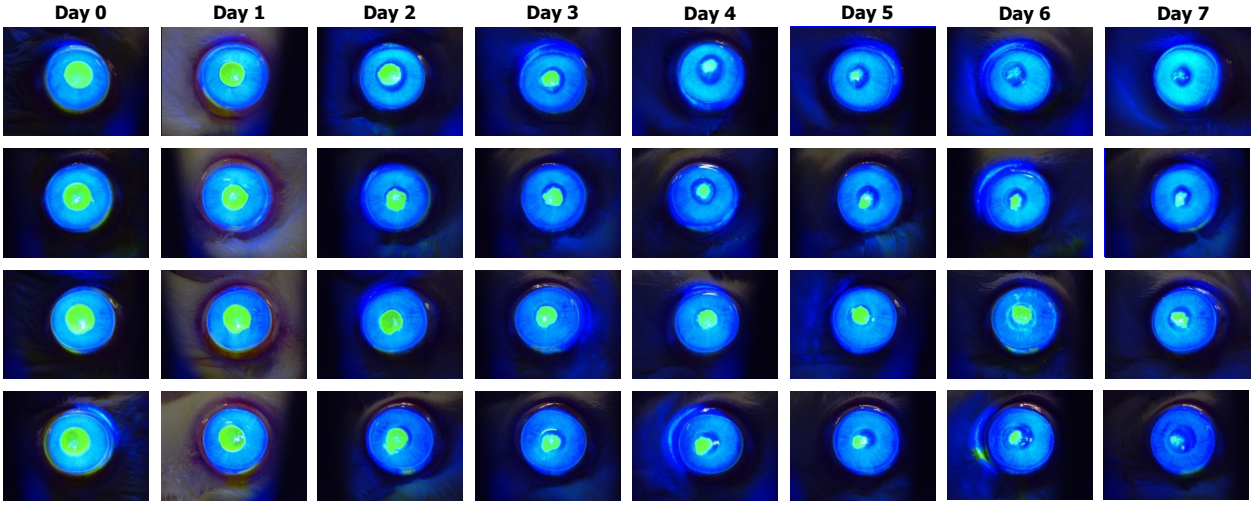


Fig. S11 Fluorescent slit lamp images showing the progression of wound healing after topical treatment with PBS. Re-epithelialization was ascertained by staining of the injured area with the topical instillation of FITC at 4 times/day.

Fig. S12

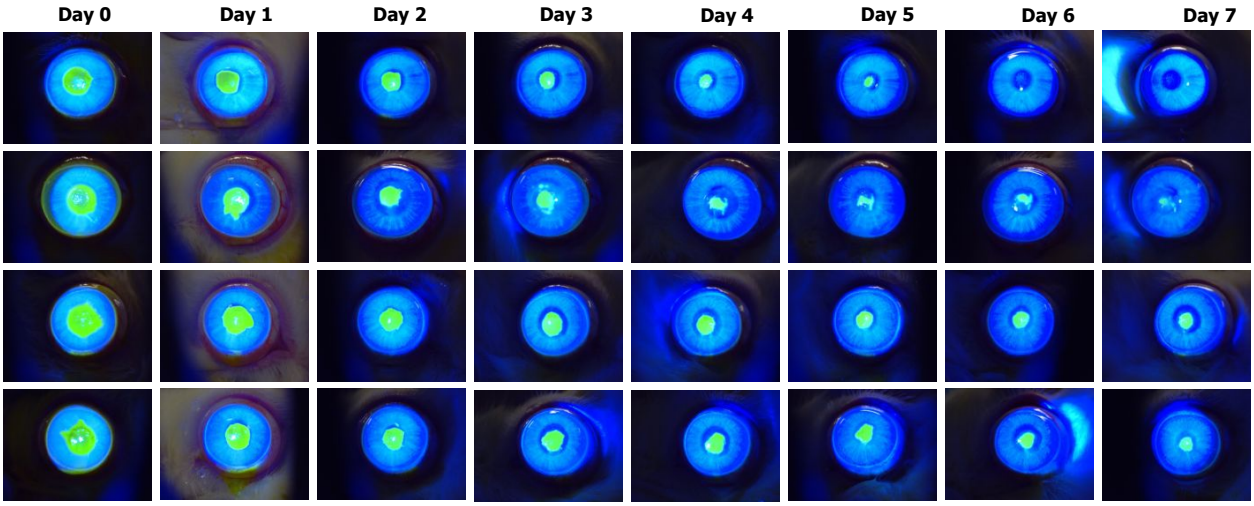


Fig. S12 Fluorescent slit lamp images showing the progression of wound healing after topical treatment with 1% εPL. The polymer solution in PBS (pH 7.0) was instilled topically at 4 times/day.

Fig. S13

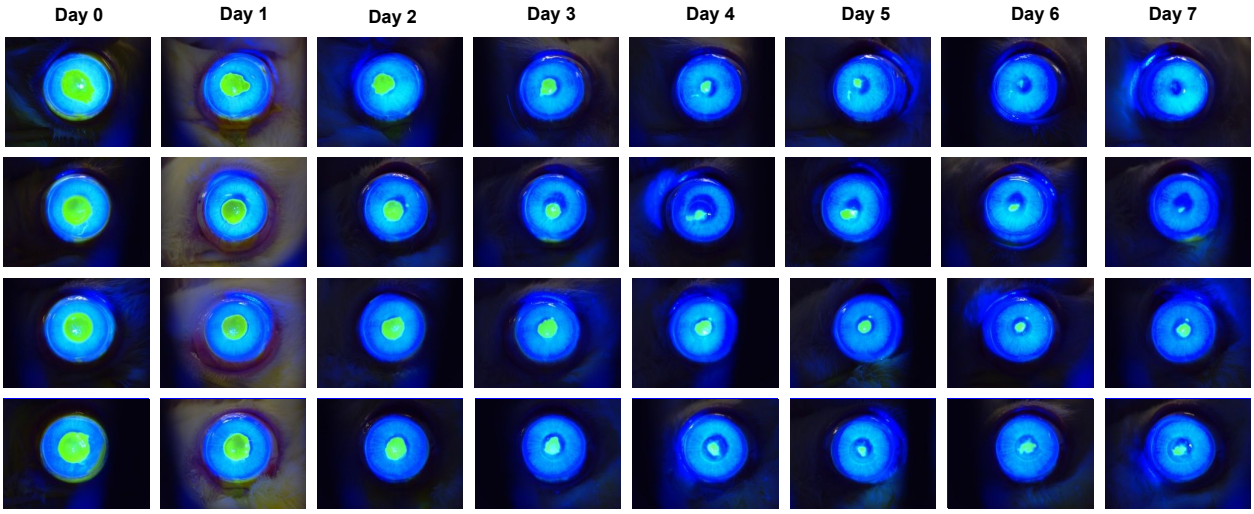


Fig. S13 Fluorescent slit lamp images showing the progression of wound healing after topical treatment with 2% εPL. The polymer solution in PBS (pH 7.0) was instilled topically at 4 times/day.

Fig. S14

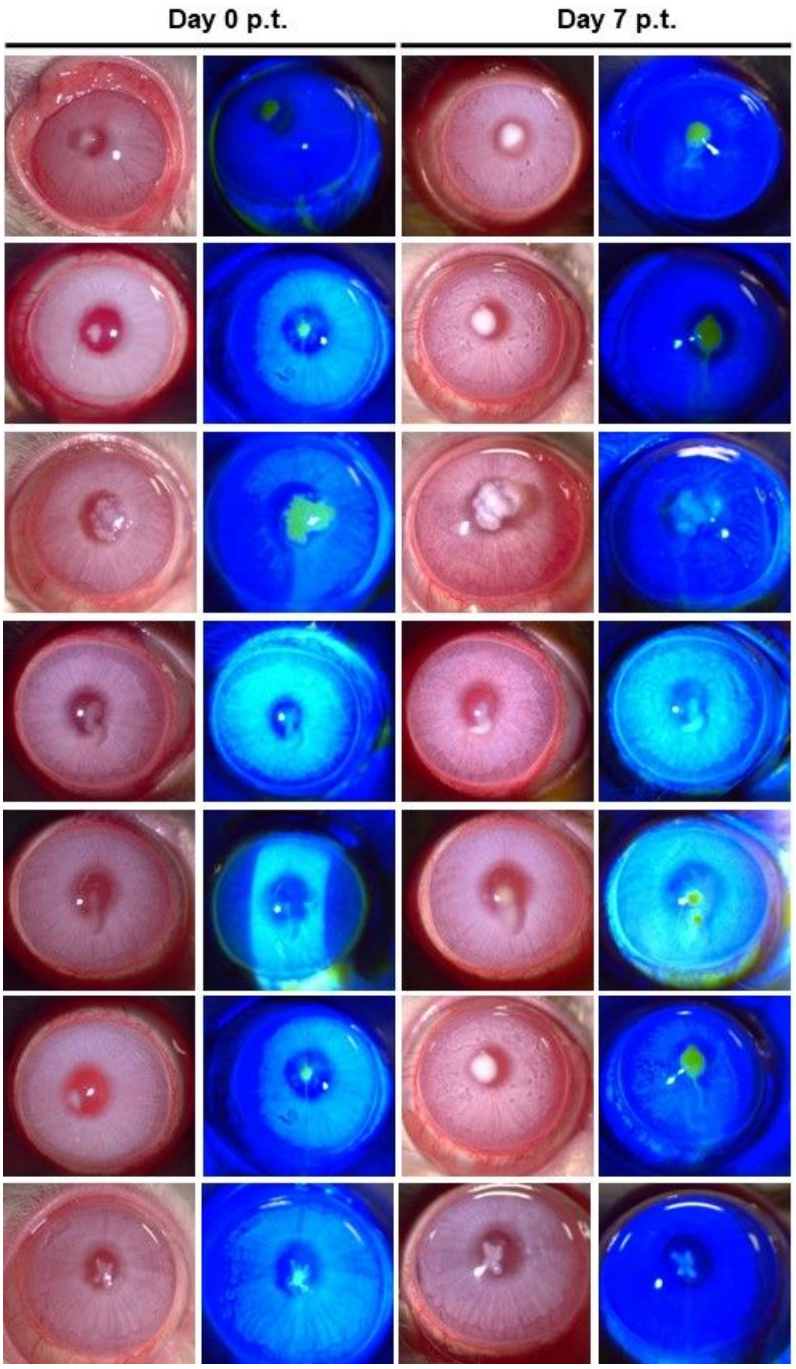


Fig. S14 SLPs and fSLPs showing the progression of the *F. solani* infected cornea after treatment with PBS (pH 7.0). PBS (pH 7.0) was instilled topically at 4 times/day.

Fig. S15

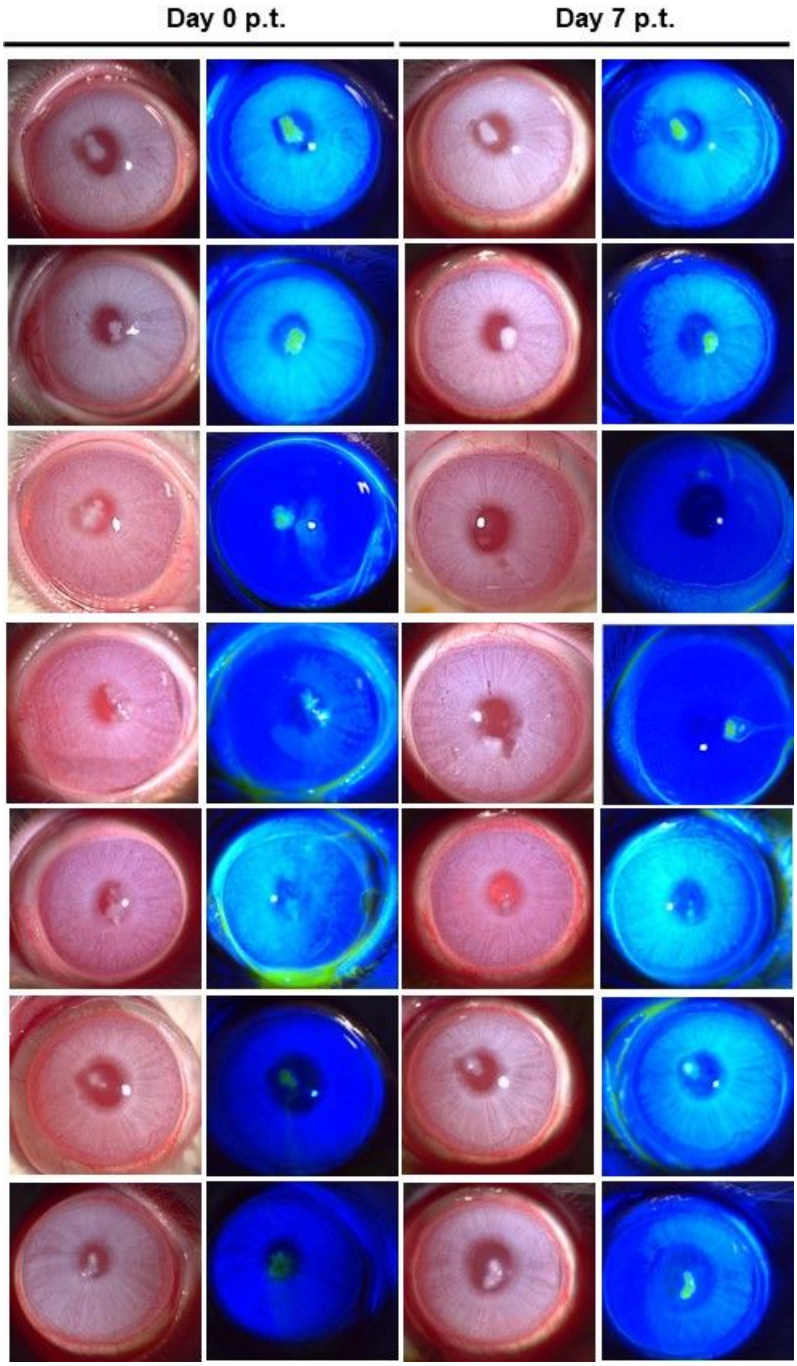


Fig. S15 SLPs and fSLPs showing the progression of the *F. solani* infected cornea after treatment with 2% εPL (pH 7.0). The polymer in PBS (pH 7.0) was instilled topically at 4 times/day.

Fig. S16

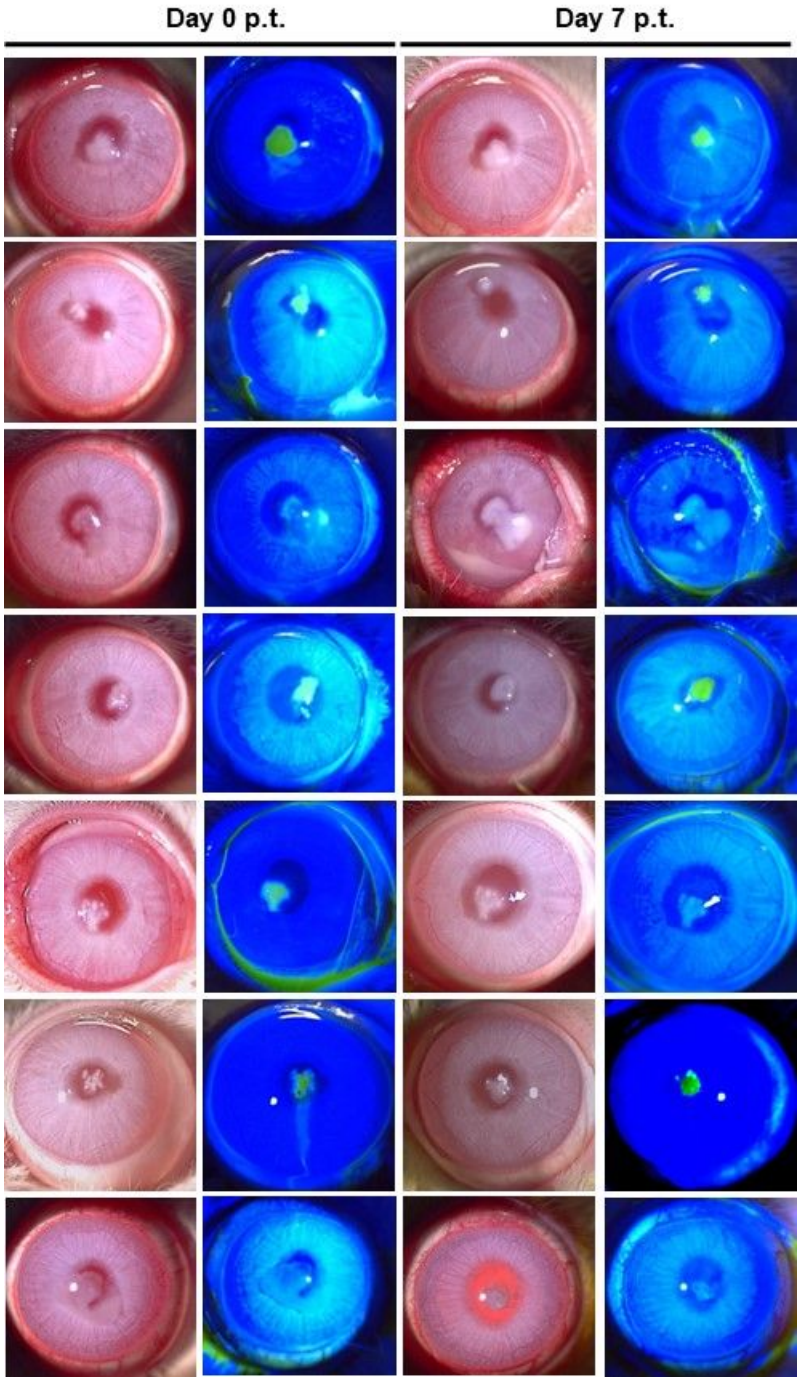


Fig. S16 SLPs and fSLPs showing the progression of the *F. solani* infected cornea after treatment with natamycin. The commercially available NATACYN® suspension was instilled topically at 4 times/day.

Fig. S17

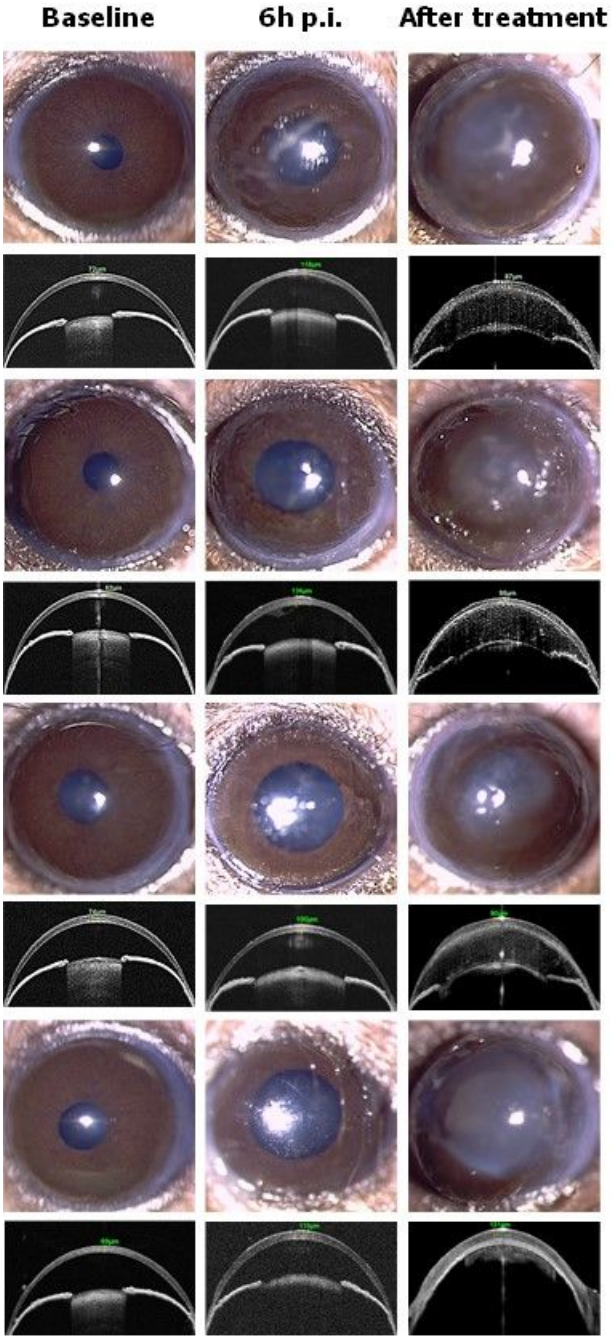


Fig. S17 Representative SLPs (top) and AS-OCT (bottom) images showing the efficacy of 2.0% εPL in a *Candida* keratitis model.